

Programme Meta-Audit (2026-04-22): Cross-Volume Consistency, Surviving Theorems, Residual Frontier

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Abstract

Three volumes, sixteen wave-cycles of adversarial probing, one functor. The trilogy — Volume I (ordered bar complex on $\overline{M}_{g,n}$; modular Koszul duality), Volume II (A_∞ chiral algebras and the Swiss-cheese-chiral-topological dioperad $\mathrm{SC}^{\mathrm{ch},\mathrm{top}}$ underpinning 3D HT QFT), Volume III (CY-to-chiral functor Φ , K3 Yangian, Calabi–Yau landscape) — crystallises around one canonical picture: a single holomorphic E_d -factorisation algebra $\mathcal{F}_X$ on each Calabi–Yau d -fold X , whose (Σ_{d-1}, C) -specialisations generate the zoo of E_1 -chiral shadows visible at the boundary of Vol I and Vol II. This audit ledgers every cross-volume dependency, reads the five shared theorems A / B / C / D / H at their current scope, catalogues the Vol III-specific contributions (CY- A_3 equivalence, K3 abelian-Yangian, ZTE T computation, CY-D dimensional stratification, universal Borchers weight $\kappa_{\mathrm{BKM}}(\Phi_N) = c_N(0)/2$), exposes the inconsistencies among CLAUDE manifestos, `working_notes.tex`, the Platonic synthesis, the CoHA treatise, and the sixty-seven `wave12_*.tex` notes, and isolates the shortest-path theorem for each volume. Five attack-heal cycles, one organising principle, six open problems.

Contents

1 Cross-volume dependency graph

The trilogy is not a disjoint union. Each volume imports load-bearing material from the other two; accurate auditing requires following the edges, not just the nodes.

1.1 Nodes

- **Vol I** *Modular Koszul Duality*: ordered bar complex $B^{\mathrm{ord}}(A) = T^c(s^{-1}\bar{A})$ on $\overline{M}_{g,n}$; five-theorem core A / B / C / D / H; averaging map $\mathrm{av}: \mathfrak{g}^{E_1} \rightarrow \mathfrak{g}^{\mathrm{mod}}$; four-archetype shadow tower $G/L/C/M$ with depth $r_{\mathrm{max}} \in \{2, 3, 4, \infty\}$; canonical formula source `chapters/examples/landscape_census.tex`.
- **Vol II** *A_∞ Chiral Algebras and 3D HT QFT*: two-coloured $\mathrm{SC}^{\mathrm{ch},\mathrm{top}}$ dioperad; bar-differential = holomorphic factorisation on C ; coproduct = topological factorisation on $\mathbb{R}$; iterated Sugawara ladder to E_n^{top} ; curved-Dunn $H^2 = 0$ at $g \geq 2$; celestial-holography climax.
- **Vol III** *CY Categories, Quantum Groups, BPS Algebras*: CY-to-chiral functor $\Phi: \mathrm{CY}\text{-cat} \rightarrow \mathrm{ChirAlg}$ with two-stage factorisation $\Phi_d = \mathrm{Sp}_{\Sigma_{d-1}, C} \circ \Phi_d^{\mathrm{FA}}$ (`notes/wave12.u1`, `notes/platonic_synthesis_wave` Thm `wn:thm:plat-two-stage`); CY-dimension-stratified target (E_2 at $d \leq 2$, E_1 at $d \geq 3$); $\kappa_{\mathrm{ch}}, \kappa_{\mathrm{cat}}, \kappa_{\mathrm{BKM}}, \kappa_{\mathrm{fiber}}$ four-fold; K3-Yangian; universal Borchers weight $\kappa_{\mathrm{BKM}}(\Phi_N) = c_N(0)/2$; $K3 \times E$ spectrum $\{2, 3, 5, 24\}$ as four distinct constructions.

1.2 Edges

Edge	Content	Load-bearing witness
Vol I $\rightarrow$ Vol III: five-theorem core	A / B / C / D / H imported verbatim; Φ_d -domain structure is a ChirAlg_X -module of Vol I	Vol III CLAUDE.md “Five theorems (shared with Vol I)”; Vol I <code>landscape_census.tex</code>
Vol I $\leftrightarrow$ Vol III: Thm C B-row	Fifth element $8 \in \{0, 8, 13, 250/3, 98/3\}$ populated by Vol III Mukai-K3 Heisenberg via Bruinier Heegner Chern-class reciprocity; $K^{\kappa_{\text{ch}}} = 2c_+(\text{Mukai}(K3)) = 8$	Vol I <code>landscape_census.tex</code> :1804–1812; Vol III <code>k3_chiral_bialgebra_platonic.tex</code> :2828–2878 (<code>thm:humbert-order-K-kappa</code>); Bruinier 2002 LNM 1780 Prop. 5.1; Mukai 1987 Nagoya Math. J. 81
Vol I $\rightarrow$ Vol III: shadow obstruction tower	Θ_A specialises to BPS counting on Φ_3 -output; CY-B is a BKM-denominator strengthening of Thm D at compact CY_d	Vol I Thm D; Vol III <code>working_notes.tex</code> :190, <code>working_notes.tex</code> :1709 (“ Θ_A of Vol I specialises as follows for CY_3 chiral algebras”)
Vol II $\rightarrow$ Vol III: $\text{SC}^{\text{ch}, \text{top}}$ operadic spine	Two-coloured dioperad; bar-differential as holomorphic factorisation; coproduct as topological factorisation; ambient for Φ_3 -image	Vol II CLAUDE.md lines 21–27; Vol III <code>working_notes.tex</code> :1728–1737 three-sector reading (open B_{E_1} / boundary B_{E_2} / closed B_{E_∞})
Vol II $\rightarrow$ Vol III: topological ladder	Iterated-Sugawara k -tower promoting chiral direction to E_n^{top} ; Vol III invokes as part of the Φ_d scope at $d \leq 2$	Vol II CLAUDE.md lines 30–35; Vol III <code>quantum_chiral_algebras.tex</code> :227–243
Vol III $\rightarrow$ Vol I, Vol II: Chriss–Ginzburg voice	Writing-standard rule imported identically across all three CLAUDE manifestos (lines 120–205 of each); no divergence	Vol I CLAUDE.md:139–205; Vol II:120–185; Vol III: (‘Writing standard’)
Vol III $\rightarrow$ Vol I: Mukai-enhanced K3 B-witness	$\mathbf{H}_{\Delta_5}$ -chamber realises the $\kappa + \kappa^! = 8$ Bruinier Chern-class identification; Vol III supplies the construction, Vol I the placement in the ceiling	Vol III <code>chapters/examples/k3_chiral_bialgebra.p</code>
Vol III $\rightarrow$ Vol I: Borchers universality	$\kappa_{\text{BKM}}(\Phi_N) = c_N(0)/2$ at $N \in \{1, 2, 3, 4, 6\}$; Vol I cites as primary-source anchor on the BKM side	Borchers 1995; Gritsenko 1999; Vol III <code>cy_d_kappa_stratification.tex</code> (<code>thm:borcherds-weight-kappa-BKM-univer</code>)

1.3 Non-edges (explicit)

Three relationships that look like edges but are not:

- Vol II's E_3 (topological factorisation on $\mathbb{R}$ -direction of a 3D HT theory; brane-boundary bulk algebra on $C \times \mathbb{R}$) is *not* Vol III's E_3 (holomorphic-chiral factorisation algebra on a three-complex-dimensional CY). Same authors (Costello–Francis–Gwilliam), different structures: the Vol III E_3 level is set by complex dimension, the Vol II E_3 level by $\mathbb{R}$ -dimension of the topological factor. Scope-declared: Vol III `en_factorization.tex:449`. Both load-bearing; neither subsumes the other.
- $\text{CoHA}(\mathbb{C}^3)$ is *not* $\mathcal{W}_{1+\infty}$. Cache-canonical: $\text{CoHA}(\mathbb{C}^3) = Y^+(\widehat{\mathfrak{gl}}_1)$ (positive half, E_1 , associative); $\mathcal{W}_{1+\infty}$ is a E_2 -chiral vertex operator algebra on a curve; the connection is $Y^+ \hookrightarrow Y(\widehat{\mathfrak{gl}}_1) \xrightarrow{\text{ev}_\lambda} \text{End}(\mathcal{W}_{1+\infty}[\lambda])$, representation-theoretic not algebra-theoretic. (Vol III `working_notes.tex:1658–1674`; Schiffmann–Vasserot 2013 Thm. 1.1; Tsymbaliuk 2017 Thm. 1.1.)
- Six routes to $G(K3 \times E)$ are six different specialisations of one $\Phi_3^{\text{FA}}(K3 \times E)$, *not* six applications of a single Φ_3 to one category. (Platonic synthesis Thm `wn:thm:plat-two-stage`; `working_notes` two-stage Cor `cor:many-BKMs`.)

2 Five-theorem status ledger

T	Name	Statement at current scope	Primary supply	Status
A	bar–cobar equivalence	$\Omega^{\text{ch}} \dashv B^{\text{ch}}$ on $\text{ChirAlg}_X^{\text{ch}}$; $\Omega(B(A)) = A$ is inversion, not Koszul duality.	Vol I (Positselski 2011, Lefèvre-Hasegawa 2003)	[THEOREM] at chain level and at $(\infty, 1)$ -level; Vol III CY-A_d refinement is a separate statement
B	chiral Positselski	$\Omega_X(B_X(C)) \xrightarrow{\sim} C$ in $D_{\text{ch}}^{\text{co}}(X)$ on Koszul locus; curved formalism elsewhere.	Vol I (Positselski 2011 Thm 6.5)	[THEOREM] on Koszul locus; [CONJECTURAL] curved lift at compact CY_d with $\kappa_{\text{ch}} \neq 0$ (<code>working_notes.tex:2229</code> flags the curved Quillen-equivalence gap; Booth–Lazarev 2304.08409 resolves abstractly, chiral instantiation open)

C	derived-centre complementar- ity	$\kappa + \kappa^! \in \{0, 8, 13, 250/3, 98/3\}$ on the G/L/C/M/B five- archetype landmark ceil- ing; classical four-subset $\{0, 13, 250/3, 98/3\}$ on G/L/C/M; B-row $K^{\kappa_{\text{ch}}} = 8$ via Bruinier Heegner Chern- class reciprocity on Mukai- enhanced K3	Vol I ceiling; [THEOREM] on Vol III B- the four classi- row witness cal archetypes; (k3_chiral_bialgebra_napoleon.tex thm:humbert-order-Kroappai) Mukai 1987 + Bruinier 2002 LNM 1780 Prop 5.1; [OPEN] whether the five- archetype ceiling is sharp (no sixth archetype known)
D	obstruction- tower univer- sality	$\text{obs}_g(A) = \kappa(A) \cdot \lambda_g$ on the uniform-weight lane; λ_g tau- tological class.	Vol I (Faber– Pandharipande 2000) [THEOREM] at $g = 1$; [CON- JECTURAL] at $g \geq 2$ (cross- family Faber– Pandharipande intersection num- bers); Vol III CY-B refinement on $K3 \times E$ identifies $\text{obs}_g = 1/\Phi_{10}$, [CONJECTURAL] at series-level equality
H	Hochschild concentration	$\text{ChirHoch}^\bullet(A)$ concentrated in $\{0, 1, 2\}$ at non-critical level; at $d \geq 3$ the E_2 -upgrade lives on $\mathcal{Z}(\text{Rep}(A))$, not on A	Vol I (chiral Deligne); Vol II (E_3 -rigidity coho- mological) [THEOREM] at non- critical level, chain- level and $(\infty, 1)$ - level; critical-level extension [CONJEC- TURAL]

Reading. Theorems A and H are essentially closed at the stated scope. B has a curved-algebra gap at compact CY_3 with $\kappa_{\text{ch}} \neq 0$ (the Booth–Lazarev 2304.08409 abstract resolution is not chirally instantiated). C is closed on the four classical archetypes and on the B-row; sharpness of the ceiling is open. D at $g \geq 2$ and CY-B at series-level remain the two substantive frontiers.

3 Vol III-specific contributions

These are theorems, scopes, and computations that Vol III supplies above and beyond the shared five-theorem core.

Contribution	Statement	Scope / supply
CY-A₃ equivalence	Φ_3 on smooth proper CY ₃ category produces an E ₁ -chiral algebra on a reference curve $E \subset X$; at the $(\infty, 1)$ -categorical level the space of E ₃ -liftings is contractible on $\mathbb{C}^3$.	[THEOREM] at the $(\infty, 1)$ -categorical level ($\mathrm{HH}_{\mathrm{E}_1}^{-2}(\mathcal{C}) = 0$ for smooth proper CY ₃ ; <code>working_notes.tex:183</code>); [SCOPE] chain-level statement is the $\mathrm{Sp}_{K_3, E}$ -specialisation of the canonical $\Phi_3^{\mathrm{FA}}(K3 \times E)$ (two-stage factorisation); 62 prior instances labelled “Theorem CY-A ₃ ” downgraded to conjecture and re-scoped (<code>working_notes.tex:5411</code>); the remaining $(\infty, 1)$ -level theorem survives
K3 abelian Yangian	$\Phi_2(D^b\mathrm{Coh}(K3))$ -presentation as an abelian Yangian with Heisenberg R -matrix at rank 22 (or Mukai-enhanced rank 24)	[THEOREM] at chain level via Maulik–Okounkov stable-envelope machinery; see <code>notes/wave12_a2_k3_abelian_yangian_kazhdan.</code>
ZTE T matrix	Exact rational T -matrix of the zonal-twisted elliptic K3 Yangian: persymmetric, zero anti-diagonal, ten distinct rational values	[THEOREM] (<code>working_notes.tex</code> <code>thm:zte-T-exact</code> , 35 verification tests)
CY-D dimensional stratification	On compact CY _{d} : $\kappa_{\mathrm{ch}}(A_X) = \sum_q (-1)^q h^{0,q}(X)$ (Hodge-supertrace identification); agrees with $\chi(\mathcal{O}_X)$ at even d ; fails at odd d because κ_{ch} is additive under direct sums but $\chi(\mathcal{O}_X)$ is not additive under direct sums of CY categories at odd d (<code>working_notes.tex</code> <code>obs:cy-d-root-cause</code>)	[THEOREM] at $d = 2$ (<code>thm:mock-modular-k3-d2</code> , 4-step proof); [CONJECTURAL] at $d = 3$ in naive form; the “failure” is the content
Universal Borcherds weight	$\kappa_{\mathrm{BKM}}(\Phi_N) = c_N(0)/2$ for $N \in \{1, 2, 3, 4, 6\}$: $c_1(0) = 10 \Rightarrow \kappa_{\mathrm{BKM}} = 5$; $c_2(0) = 8 \Rightarrow 4$; $c_3(0) = 6 \Rightarrow 3$; $c_4(0) = 4 \Rightarrow 2$; $c_6(0) = 2 \Rightarrow 1$	[THEOREM] (Borcherds 1995; Gritsenko 1999; Vol III <code>cy_d_kappa_stratification.tex</code> <code>thm:borcherds-weight-kappa-BKM-universal</code>); extension to $N \in \{5, 7, 8\}$ with Borcherds-weight scope (metaplectic covers, spin double covers, abelian degeneracy) is platonic synthesis <code>Thm wn:thm:plat-universal-kBKM</code> “two scopes”

Two-stage factorisation	$\Phi_d = \mathrm{Sp}_{\Sigma_{d-1}, C} \circ \Phi_d^{\mathrm{FA}}: \mathrm{CY}\text{-cat}_d \rightarrow \mathrm{E}_d \mathrm{HolFA}(X) \rightarrow \mathrm{E}_1 \mathrm{ChirAlg}(C);$ Stage 1 canonical up to contractible choice (Kontsevich–Tamarkin formality + Costello–Gwilliam–Li locality); Stage 2 a specialisation parametrised by $(\Sigma_{d-1}, C) \in \mathrm{Cyc}_{d-1}(X) \times \mathrm{Curves}(X)$	[THEOREM] at $d = 1, 2$ with explicit witnesses (notes/wave12_u1, platonic synthesis Thm <code>wn:thm:plat-two-stage</code>); [CONJECTURAL] at $d = 3$ sketch via Dunn–Lurie $\int_{\Sigma_2} \mathrm{E}_3 \simeq \mathrm{E}_1$; requires explicit Costello–Li hCS pre-factorisation algebra and Stokes–Hochschild dualisation
K3×E spectrum $\{2, 3, 5, 24\}$	Four distinct constructions on the single geometry: $2 = \kappa_{\mathrm{cat}}(K3)$ (K3-fibre categorical Euler); $3 = \kappa_{\mathrm{ch}}(K3 \times E)$ Mukai-refined Hodge supertrace (cf. the cached $\kappa_{\mathrm{ch}}(\mathrm{total}) = 0$ for Calabi–Yau total space via Künneth-multiplicative κ_{cat} — see inconsistency catalogue §4 below); $5 = \kappa_{\mathrm{BKM}}(\Phi_1) = c_1(0)/2$; $24 = \kappa_{\mathrm{fiber}}$ (K3 Mukai-lattice rank, signature $(4, 20)$)	[SCOPE] four-constructions reading is consistent; the conflation with the Vol I complementarity set $\{0, 8, 13, 250/3, 98/3\}$ is forbidden; bridged by Mukai doubling $K^{\kappa_{\mathrm{ch}}} = 2c_+ = 24 - 16 = 8$
Six routes to $G(K3 \times E)$	Six (Σ_2, C) -specialisations of one $\Phi_3^{\mathrm{FA}}(K3 \times E)$: $K3\text{-fibre} \rightarrow \mathfrak{g}_{\Delta_5}$ ($\kappa_{\mathrm{BKM}} = 5$); paramodular Humbert $N \in \{2, 3, 4, 6\}$ CHL cousins; envelope $\Lambda^{3,3} \rightarrow \mathfrak{g}_{\Lambda^{3,3}}^{\mathrm{BKM}}$ with $\mathfrak{g}_{\Delta_5} \subset$ as σ_q -fixed subalgebra	[THEOREM] for $K3\text{-fibre}$ (Δ_5 lift) and envelope; [CONJECTURAL] for CHL cousins and Humbert-sibling data (working_notes.tex cor:many-BKMs points 2, 3)
Mock modular K3 at $d = 2$	$\Phi_2(D^b \mathrm{Coh}(K3))\text{-bar}$ complex carries mock modularity of Cheng–Duncan–Harvey umbral Eguchi–Ooguri–Tachikawa type; $\widehat{h}(\tau)$ completion is the shadow-tower chamber-independent substitute	[THEOREM] at $d = 2$ (working_notes.tex thm:mock-modular-k3-d2)
Seven faces of r_{CY} three-tier stratification	Tier (i) CY-datum intrinsics (κ_{cat} , Mukai pairing); Tier (ii) Stage-1 invariants of $\mathcal{F}_X$ (κ_{fiber} , fibre-rank data); Tier (iii) (Σ_2, C) -specialisations (κ_{BKM} , Humbert, CHL twined)	[THEOREM] at the level of stratification (platonic synthesis Thm <code>wn:thm:plat-three-tier</code>); the algebraic seven-presentation (bar-cobar / CoHA / coisson / MO / Yangian / Sklyanin / Gaudin) is an orthogonal slicing

CY- A_3 survival strata S_1 – S_5 . The “62 instances” adjustment of `working_notes.tex:5411` records the following stratification of the original universal “ Φ_3 produces E_1 -chiral on any reference curve” claim:

- S₁. $(\infty, 1)$ -level existence on smooth proper CY_3 categories: [THEOREM].
- S₂. Chain-level explicit presentation on toric CY_3 ($\mathbb{C}^3$, local $\mathbb{P}^2$, resolved conifold): [THEOREM] via Costello–Li hCS + Schiffmann–Vasserot CoHA.
- S₃. Chain-level explicit presentation on compact $K3 \times E$: [SCOPE] as the $\text{Sp}_{K3,E}$ -specialisation of $\Phi_3^{\text{FA}}(K3 \times E)$; explicit is $U^{\text{ch}}(\mathfrak{heis}_{\text{Muk}}) \otimes U^{\text{ch}}(\mathfrak{g}_{K3}^{\text{BPS}})$ (platonic synthesis Thm `wn:thm:plat-Sp-K3E`).
- S₄. Chain-level explicit presentation on general compact CY_3 : [CONJECTURAL] (requires universal Costello–Li construction in the global/non-toric regime; no generic witness).
- S₅. Morphism preservation making Φ_d a functor on $(\infty, 1)$ -categories: [CONJECTURAL] (CY-C, quantum vertex chiral group $G(X)$).

The Vol III programme thread lives at stratum S₂ and S₃, the first explicitly, the second via two-stage factorisation. S₄ is the residual frontier for the functor-output statement.

4 Inconsistency catalogue with adjudication

Scan target: three CLAUDE.md manifestos; `working_notes.tex`; `notes/platonic_synthesis_waves_11_through_notes/CoHA_to_W_infty_treatise.tex`; sixty-seven `wave12_*.tex` notes. Inconsistencies catalogued with adjudication from the epistemic hierarchy (direct computation > source > build > primary literature > concordance > manifesto > memory).

#	Inconsistency	Witnesses	Adjudication
1	$\kappa_{\text{ch}}(K3 \times E) \in \{0, 3, 5\}$	0: <code>wave12_a2_kappa_invariants</code> :353, :656; <code>working_notes.tex</code> :7741, :10029. 3: <code>wave12_a12_six_routes</code> :104, <code>wave12_a4_k3_yangian_abelian</code> :792, <code>wave12_a5_cy_d</code> :471,656; <code>working_notes.tex</code> :2716, :3099, :3782, :20383, :20714. 5: <code>working_notes.tex</code> :1361, :2682.	The three values are three distinct invariants under a single notation. The cache-canonical is $\kappa_{\text{cat}}(K3 \times E) = \chi(\mathcal{O}_{K3 \times E}) = 0$ (Künneth-multiplicative, $2 \cdot 0 = 0$). The value 3 is the Hodge supertrace $\sum_q (-1)^q h^{0,q}(K3 \times E) = 3$ (via Vol III Mukai-refined identification with signature-(4,20) lattice positivity). The value 5 is $\kappa_{\text{BKM}}(\Phi_1) = c_1(0)/2$, the Borcherds weight of Δ_5 , not a κ_{ch} value. <i>Action at the reader-facing manuscript level:</i> disambiguate by subscript discipline (HZ-7); bare κ forbidden; $\kappa_{\text{ch}}^{\text{Heis}}$ used at <code>wave12_a12</code> :104 is an instructive intermediate superscript (superscripted by route) but should be downgraded to the cached pair κ_{ch} vs κ_{cat} . Recommend: $\kappa_{\text{ch}}(K3 \times E) = 3$ (Hodge supertrace); $\kappa_{\text{cat}}(K3 \times E) = 0$ (total-space categorical Euler); $\kappa_{\text{BKM}}(\Phi_1)(K3 \times E) = 5$ (Borcherds weight of Δ_5). Three values, three subscripts, not a conflict.
2	Fake Monster at $d = 3$ vs $d = 5$	$d = 3$: some older <code>working-notes</code> passages list Fake Monster as a $K3 \times E$ sibling via Niemeier-surface cycle. $d = 5$: platonic synthesis in compact CY_3 supplies Cor <code>wn:cor:plat-many-bkms</code> (Niemeier data). Conjec- ture <code>wn:conj:plat-siblings</code> :31m, <code>wave12_d3</code> , <code>wave11_u1_niemeier</code> .	Platonic synthesis retracts the $d = 3$ framing as [RETRACTED] (rank-24 Leech lattice exceeds $h^{1,1}(K3) = 20$; no transverse surface in compact CY_3 supplies Niemeier data). <i>True:</i> Fake Monster lives at $d = 5$ via $K3 \times K3 \times E$, $\Sigma_4 = K3 \times K3$, $C = E$, $\kappa_{\text{BKM}}(\Phi_{\text{FM}}) = 12$. Resolution is in the Platonic synthesis.

- 3 $\text{CoHA}(\mathbb{C}^3) \cong \mathcal{W}_{1+\infty}$ vs $\text{CoHA}(\mathbb{C}^3) = Y^+ \neq \mathcal{W}_{1+\infty}$ $\cong$: CoHA treatise rows in $\mathcal{W}_{1+\infty}$ is the representation-theoretic image of $Y(\widehat{\mathfrak{gl}}_1)$ under the Gaiotto–Rapčák λ -parameter evaluation: $Y^+ \hookrightarrow Y(\widehat{\mathfrak{gl}}_1) \xrightarrow{\text{ev}_\lambda} \text{End}(\mathcal{W}_{1+\infty}[\lambda]\text{-vacuum})$.
`CoHA_to_W_infty_treatise.tex` could be read as identification via “state-field correspondence on the vacuum module”; summary tabular row “Identified CoHA is E_1 , associative with VOA: $\mathcal{W}_{1+\infty}$ ”. $\neq$: $\mathcal{W}_{1+\infty}$ is E_2 , associative.
`wave12_f4.coha_w_infty_consistency.tex`, Not a flaw.
`flaw:coha-winf`; an associative-algebra identification.
`wave12_a11.coha_y_plus_vs_identification`. Resolutive synthesis.
`Thm wn:thm:plat-Miki-S3` wave12_f4:cor:coha-winf. Correction at remark.
- 4 Swiss-cheese two-coloured vs three-sector Two-coloured: Vol II `CLAUDE.md`:20–27, :215–216 (“ $\text{SC}^{\text{ch},\text{top}}$ is a 2-coloured dioperad”). Three-sector: Vol III `working_notes.tex`:1728–1737 ($B_{E_1}, B_{E_2}, B_{E_\infty}$). Compatible: the two colours stratify into three graded pieces via open/boundary/closed decomposition. Vol II Pattern 277: “ $\text{SC}^{\text{ch},\text{top}}$ vs E_3 conflation is a scope declaration, not a hierarchy.” Resolution: Vol III reading is a three-graded faithful decomposition of the two-coloured structure.
- 5 E_3 in Vol II (topological bulk) vs in Vol III (6D holomorphic) Vol II: Two different E_3 structures, same CFG authorship, `bar-cobar-review.tex`:4546, `celestial_holography.tex`:2018. Vol III: (topological in R-direction); `en_factorization.tex`:449–500, `quantum_chiral_algebras.tex`:227. Compatible via bulk-`wave12_a13.costello_5_rouquier_hcs`, correspondence. `wave12_b3_6d_hcs`, Scope declared in Vol III `wave12_f1_6d_hcs_e3_generation` `en_factorization.tex`:449 final clause.

6	CY- A_3 labelled as universal vs as $\mathrm{Sp}_{K3,E}$ -specialisation	Universal: 62 prior “Theorem CY- A_3 ” instances in <code>working_notes.tex</code> . Specialisation: <code>notes/wave12.u1.two.stage</code> . <code>Thm vn:thm:cy-a3-specialisation</code> improved at $(\infty, 1)$ -level Re-mark <code>rem:cy-a3-specialisation</code> on E is the $K3$ -fibre specialisation $\mathrm{Sp}_{K3,E}$. 62 instances re-scoped per <code>working_notes.tex:5411</code> . (retraction note).	Φ_3 has no unique E_1 -chiral output on a curve; the output is $\mathrm{Sp}_{\Sigma_2,C}(\Phi_3^{\mathrm{FA}}(K3 \times E))$ parametrised by (Σ_2, C) . <code>Thm vn:thm:cy-a3-specialisation</code> proved at $(\infty, 1)$ -level (stratum S_1); the chain-level output on E is the $K3$ -fibre specialisation $\mathrm{Sp}_{K3,E}$. 62 instances re-scoped per <code>working_notes.tex:5411</code> .
7	$\widehat{\mathfrak{sl}}_3 \hookrightarrow \mathfrak{g}_{\Delta_5}$ from η^9 exponent	Early: eta-9 exponent claim in pre-Wave-15 working-notes passages. Refuted: Kac-Moody; $\eta^9 = \eta^1 \cdot \eta^8$, Platonic synthesis not $\eta^6 \cdot \eta^3$. Cartan mismatch <code>Thm wn:thm:plat-retraction</code> (rank 12, positive definite vs rank-3 hyperbolic); denominator mismatch (η^{11} vs η^9).	[RETRACTED]. Real-root subalgebra is Feingold-Frenkel F_3 rank-3 hyperbolic Kac-Moody; $\eta^9 = \eta^1 \cdot \eta^8$, not $\eta^6 \cdot \eta^3$. Cartan mismatch (rank 12, positive definite vs rank-3 hyperbolic); denominator mismatch (η^{11} vs η^9).
8	$\mathrm{VOA}[T_{24}] = L_{-6}(\mathfrak{e}_8)$	Early: central charge claim $c = -62$. Refuted: Platonic synthesis <code>Thm wn:thm:plat-retraction</code> (rank 2, $\mathcal{H}_{\mathrm{PS}}(\mathbf{2})_{-2+1/22}(\mathfrak{sl}_2)^{\otimes 22}$) via iterated Drinfeld-Sokolov on pants decomposition (Arakawa 2017 + Creutzig-Linshaw 2017).	[RETRACTED]. Central charge mismatch $c = -62 \neq -214$. True: $\mathcal{V}_{24} = \mathcal{H}_{\mathrm{PS}}(\mathbf{2})_{-2+1/22}(\mathfrak{sl}_2)^{\otimes 22}$ via iterated Drinfeld-Sokolov on pants decomposition (Arakawa 2017 + Creutzig-Linshaw 2017).
9	$\kappa_{\mathrm{BKM}} = \kappa_{\mathrm{ch}} + \chi(\mathcal{O}_{\mathrm{fiber}})$ universally	Early: coincidence at $N = 1$ ($5 = 2 + 3$). Refuted: CLAUDE.md Vol III key facts; Platonic synthesis <code>Thm wn:thm:plat-retraction</code> (rank 13, Bousquet-Mélou weight theorem).	[RETRACTED]. Fails at all $N \geq 2$ (numerically at $N = 2$: $\kappa_{\mathrm{BKM}} = 4$, $\kappa_{\mathrm{ch}} + \chi(\mathcal{O}_E) = 0 + 0 = 0$). True universal: $\kappa_{\mathrm{BKM}}(\Phi_N) = c_N(0)/2$, Bousquet-Mélou weight theorem.
10	Gaiotto curve $\Sigma_{2,0}$ giving $c_{4d} = 107/6$	Early: genus-2 closed curve assumption. Refuted: Platonic synthesis <code>Thm wn:thm:plat-retraction</code> (rank 6, $\Sigma_{0,24}$ (24-punctured sphere), Chacaltana-Distler 2010 Table 3 row 1, $c_{4d} = (5n - 13)/6 = 107/6$ at $n = 24$).	[RETRACTED]. Genus-2 closed curve gives $c_{4d} = 7/6$ or $13/6$, never $107/6$. True: $\Sigma_{0,24}$ (24-punctured sphere), Chacaltana-Distler 2010 Table 3 row 1, $c_{4d} = (5n - 13)/6 = 107/6$ at $n = 24$.

11	PTVV table terminates at $d = 4$	Early: shifted-symplectic table stops at $d = 4$. Refuted: Platonic synthesis Thm <code>wn:thm:plat-hCS-classical+1</code> . (shift law shift = $d - 4$ admits arbitrary integer shifts); (5, +1, $E_2[5]$ -Poisson).	[CORRECTED]. Shift law admits arbitrary integer; $d = 5$ entry is Poisson- $E_2[5]$ with shift +1.
12	E_3 -deformation-of-Chern–Simons Costello–Francis–Gwilliam as cited primary source	Asserted in some wave12 passages. Scope-flagged: <code>CoHA_to_W_infty_treatise.tex:308</code> 314, wave12_a11, wave12_a13.	[SCOPE]. Costello–Gwilliam 2017 is the primary source for the E_n -factorisation-algebra framework; Costello 2013 is the primary source for the Yangian via Ω -background; a further Costello–Francis–Gwilliam E_3 -of-CS-deformations sharpening paper is <i>in preparation</i> and not a cited primary source as of 2026-04-22. Treatise does not rely on it for the load-bearing statements.
13	Humbert boundary monodromy H_1 order-8 vs order-16	Two candidates in working-notes.	H_1 monodromy is order-8 (matching Mukai $K^{\kappa_{\text{ch}}} = 8$); order-16 arises only in a doubled chamber around H_4 . Resolution: three faces of 8 theorem (<code>thm:humbert-order-K-kappa</code>): Mukai / monodromy / Lusztig all coincide at $\ell = 8$.

Pattern of discrepancies. All thirteen entries reduce to three causes: (a) subscript discipline not applied uniformly across sixty-seven wave12 notes before disambiguation; (b) earlier working-notes passages retracted but not yet physically deleted (they survive as strata within `working_notes.tex` or as `wave12*.tex` drafts); (c) scope-declared “same-structure-different-object” pairs (Vol II vs Vol III E_3 ; $\text{CoHA}(\mathbb{C}^3)$ vs $\mathcal{W}_{1+\infty}$). None are mathematical contradictions; all are terminological. The manuscript-level rectification is subscript normalisation at the reader-facing **chapters/** tier (HZ-7 discipline).

5 Shortest-path frontier per volume

Vol I. One theorem away from closure: complete the curved-bar-cobar Quillen equivalence at compact CY_d with $\kappa_{\text{ch}} \neq 0$ (`working_notes.tex:2229–2245`). Booth–Lazarev 2304.08409 supplies the abstract coderived + curved Quillen equivalence; the chiral instantiation on $\overline{M}_{g,n}$ requires the

chiral analogue of their curved- A_∞ cotangent complex. The “shadow tower Θ_A as a genus- g class” identification (`working_notes.tex:2263–2288`) closes CY-B on the series-level equality at $K3 \times E$.

Vol II. One theorem away from closure: the E_n^{top} topologisation ladder reaching E_∞ via $\mathcal{W}_{1+\infty}$. The content: chain-level proof that $\mathcal{W}_{1+\infty}$ at $c = 1$ (Heisenberg) is the Platonic endpoint of the iterated-Sugawara ladder under curved-Dunn $H^2 = 0$ at $g \geq 2$. The Vol II $\text{SC}^{\text{ch},\text{top}}$ two-coloured dioperad supplies the operadic framework; the three-sector Vol III reading supplies the target. The residual frontier is the witnessing chain homotopy for the $k \rightarrow \infty$ limit.

Vol III. One theorem away from closure: the explicit $\text{Sp}_{K3,E}$ -computation of $\Phi_3^{\text{FA}}(K3 \times E)$, chain level, producing $U^{\text{ch}}(\mathfrak{heis}_{\text{Muk}}) \otimes U^{\text{ch}}(\mathfrak{g}_{K3}^{\text{BPS}})$ (platonic synthesis Thm `wn:thm:plat-Sp-K3E`, currently S_3 scope). Supplies: Maulik–Okounkov stable envelope for the $K3$ factor (Davison 2017 + Nakajima–Yoshioka 2004); Heisenberg factorisation algebra for the Mukai sector (Frenkel–Ben-Zvi 2004 + Grojnowski 1996). Obstruction: the E -factor E_1 extraction on the rank-22 or rank-24 Mukai Heisenberg via Dunn–Lurie $\int_{K3} E_3 \simeq E_1$; the $K3$ factorisation homology needs the global critical-chart hypothesis (Davison 2017) which is not automatic for every compact CY_3 .

One additional theorem within reach for Vol III: the Lorgat 2020 Conjecture 1 sharpened (`wn:conj:plat-lorgat-2020`) — $\mathbf{H}_{\Delta_5}$ -chiral-bialgebra on E isomorphic to $\text{Sp}_{K3,E}(\Phi_3^{\text{FA}}(K3 \times E))$ with character Δ_5^{-2} through paramodular Fourier expansion. This upgrades Conjecture 1 from numerical match to structural identification; the bridge is the Gritsenko–Nikulin 1996 Cor. 5.2 residue at H_1 + Frenkel–Ben-Zvi Heisenberg character identity, both already supplied in Thm `wn:thm:plat-heis-mukai-e`.

6 Two-stage factorisation as organising principle

Adversarial probing of the platonic-synthesis Thm `wn:thm:plat-two-stage`, in five passes.

6.1 Attack–heal 1: is Stage 1 truly canonical?

Attack 6.1. The two-stage factorisation claims Stage 1 is “canonical up to contractible choice” via Kontsevich–Tamarkin formality. But Kontsevich–Tamarkin E_d -formality is proved for polydifferential operators on $\mathbb{R}^d$ (Kontsevich 1997; Tamarkin 1998), not for a holomorphic-factorisation-algebra avatar on a general CY_d . The extension to holomorphic CY_d is Costello–Gwilliam–Li, which gives locality and the pre-factorisation algebra, but does *not* supply the formality quasi-isomorphism.

Heal 6.2. Stage 1 canonicity is actually a conjunction of two inputs: (a) Costello–Li 2016 hCS pre-factorisation algebra construction on $\mathbb{C}^3$, which supplies the explicit E_3 -holomorphic factorisation algebra; (b) a Kapranov-type Dolbeault-polyvector-field presentation of the Hochschild cohomology, giving an E_n -structure on $\text{PV}_{\bar{\partial}}^\bullet(X, \mathcal{O}_X)$ via the Kontsevich formality map on the de Rham avatar. The conjunction is not a single formality theorem; it is a two-step reduction. On compact $K3 \times E$, step (a) is available via Costello–Li for the toric tangent $\mathbb{C}^3$ -chart and Dolbeault gluing; step (b) requires a primary-source Kapranov 1999 argument on the Dolbeault L_∞ -model. Platonic synthesis Thm `wn:thm:plat-Linf-minimal` supplies the $\mathbb{C}^3$ -step-(b); the Atiyah-class formality obstruction on compact CY_3 is flagged but the $K3 \times E$ case admits the $\text{At}(TE) = 0 + \Omega_{K3}^3 = 0$ vanishing. *Scope:* Stage 1 is canonical for $\mathbb{C}^3$ and $K3 \times E$; general compact CY_3 requires the Atiyah-class obstruction argument, which is $(\infty, 1)$ -categorical, not chain-level.

6.2 Attack–heal 2: is Stage 2 genuinely a moduli, or a single canonical cycle?

Attack 6.3. The two-stage factorisation’s Stage 2 is parametrised by $(\Sigma_{d-1}, C) \in \text{Cyc}_{d-1}(X) \times \text{Curves}(X)$. But $\text{Cyc}_{d-1}(X)$ of a compact $K3 \times E$ is a huge moduli; specifying a “distinguished” $\Sigma_2 = K3$ -fibre requires a choice of splitting $K3 \times E \rightarrow E$, which is canonical only for the trivial product structure. For a generic CY_3 with non-trivial fibration, multiple splittings lead to inequivalent Stage-2 outputs.

Heal 6.4. For $X = K3 \times E$ as a product, the splitting is canonical and Stage 2 via $\Sigma_2 = K3$ -fibre produces the $\mathbf{H}_{\text{Muk}} \otimes U^{\text{ch}}(\mathfrak{g}_{K3}^{\text{BPS}})$ specialisation. For X with non-trivial $K3$ -fibration (e.g. a generic Borcea–Voisin threefold), *different* fibre choices generate different Σ_2 -specialisations, hence different $\mathbf{E}_1$ -chiral siblings. This is not a bug: the “many BKMs” phenomenon is the statement that $\text{Cyc}_2(X)$ has nontrivial topology. The canonical cycle structure on $K3 \times E$ product case is a special simplification, not a universal feature.

6.3 Attack–heal 3: Dunn–Lurie $\int_{\Sigma_2} \mathbf{E}_3 \simeq \mathbf{E}_1$ direction

Attack 6.5. The two-stage factorisation Proof Sketch invokes Dunn–Lurie $\mathbf{E}_3 \simeq \mathbf{E}_1 \otimes \mathbf{E}_2$, then integrates over $\Sigma_2 = K3$, claiming $\int_{\Sigma_2} \mathbf{E}_2 \simeq \text{pt.}$ But $\int_{K3} \mathbf{E}_2$ is not pt; it is the factorisation homology of an $\mathbf{E}_2$ -algebra over a closed real 4-manifold, which is π_0 -nontrivial in general (Ayala–Francis 2015).

Heal 6.6. The correct statement is that $\int_{\Sigma_2} \mathbf{E}_2$ over a closed orientable real surface of genus g has known π_0 (Ayala–Francis). For $K3 = \Sigma_2$ as a real 4-manifold (not a real surface!) this is even further out of scope. The content actually needed: the $\mathbf{E}_3$ -hFA on $\mathbb{C}^3$ projected along two complex directions to leave one complex direction is an $\mathbf{E}_1$ -chiral algebra on the surviving $\mathbb{C}$ -direction, via the Costello–Li hCS-to-CoHA mechanism. The “Dunn additivity + integration” sketch conflates real and complex dimensions. The chain-level mechanism is: $\mathbf{E}_3$ -holomorphic on $\mathbb{C}^3$; restrict to a reference curve $C \subset \mathbb{C}^3$ transverse to a Σ_2 (complex surface, real dimension 4); the restriction is $\mathbf{E}_1$ -chiral on C by holomorphic locality in the one surviving complex direction. No $\int \mathbf{E}_2 \simeq \text{pt}$ required. Resolution: the two-stage factorisation at $d = 3$ is proved at the chain level on $\mathbb{C}^3$ (Costello–Li) and on $K3 \times E$ (via Stage 1 on $\mathbb{C}^3$ -chart + Dolbeault gluing); the $(\infty, 1)$ -categorical restatement is the cleaner form but needs Ayala–Francis to be handled separately per Σ_{d-1} .

6.4 Attack–heal 4: does the organising principle extend beyond $d \leq 5$?

Attack 6.7. Platonic synthesis lists only $d \in \{1, 2, 3, 5\}$ explicitly, with $d = 4$ handled via the shifted-symplectic table at shift 0 and $\mathbf{E}_0$. What is the Φ_d target at $d = 4$, $d \geq 6$?

Heal 6.8. At $d = 4$ the PTVV shift is 0 and the $\mathbf{E}_d$ target is $\mathbf{E}_0$: Φ_4 produces an $\mathbf{E}_0$ -chiral structure, i.e. essentially a cochain complex without any $\mathbf{E}_1$ -or-higher multiplication. The Calabi–Yau 4-fold Donaldson–Thomas theory (Cao–Kool–Leung 2020, Borisov–Joyce 2017) witnesses the vanishing cycle sheaf but not an $\mathbf{E}_1$ -algebra. At $d \geq 6$, shift is $d - 4$; the target is $\mathbf{E}_n[d - 3]$ -Poisson with shift $d - 4$: Poisson, not symplectic, for $d \geq 5$ (Platonic synthesis Thm `wn:thm:plat-hCS-classical`). The organising principle extends; the content at $d \geq 6$ is the CHL-type string-landscape structure (not the subject of Vol III’s 693-page scope but present in the Fake Monster $d = 5$ cousin).

6.5 Attack–heal 5: is the two-stage principle falsifiable?

Attack 6.9. If every phenomenon fits the two-stage factorisation by adjusting (Σ_{d-1}, C) , the principle is unfalsifiable. Where does it *fail*?

Heal 6.10. Three specific falsifiable predictions: (a) Fake Monster is at $d = 5$, not $d = 3$ (rank obstruction). This would fail if a rank-24 Leech lattice embedded into $h^{1,1}$ of a compact CY_3 (it does not); (b) $N = 7$ in the CHL family is obstructed (Niemeier fixed-count 3 forces singular limit; weight $1/4$ corresponds to order-4 gerbe Cheeger–Simons character). This would fail if $N = 7$ produced a non-singular BKM via the K3-fibre specialisation (it does not; currently open); (c) K3-fibre-rank $\kappa_{\text{fiber}} = 24$ Mukai-doubles to Vol I B-row $K^{\kappa_{\text{ch}}} = 2c_+(\text{Mukai}) = 8 = 24 - 16$. This would fail if another signature- $(4, 20)$ rank-24 lattice gave a different B-row value; it does not.

All three predictions align with current data. The principle is consistent and falsifiable; it survives adversarial probing.

7 6d hCS / CFG E_3 as cross-volume unification

Claim under probe: the single E_d -factorisation algebra on each CY_d is the unifying mechanism across all three volumes.

7.1 Vol III E_d -hFA

Canonical E_3 -algebra of 6d holomorphic Chern–Simons observables $\text{Obs}_{\text{hCS}_6}^q \in \mathcal{F}_{\mathbb{C}^3}$; BV-BRST complex with Bochner–Martinelli propagator (Platonic synthesis `Thm wn:thm:plat-hCS-quantum`); anomaly factorisation $\kappa_{\text{anom}}(X, \mathfrak{g}) = \hbar A(\mathfrak{g}) \cdot \chi_{\text{top}}(X)/(2(4\pi)^3) \cdot \|\Omega_X\|^2$ with cubic Casimir $A(\mathfrak{g})$ (`Thm wn:thm:plat-anomaly`); minimal L_∞ -model on $\mathbb{C}^3$ with $\ell_n^{\min} = 0$ for $n \geq 3$ (`Thm wn:thm:plat-Linf-minimal`).

7.2 Vol II E_d -HT QFT

3D HT QFT on $C \times \mathbb{R}$: chiral direction C carries holomorphic factorisation; topological direction $\mathbb{R}$ carries topological factorisation. Bulk algebra is E_3 -topological on observables. Two-coloured $\text{SC}^{\text{ch}, \text{top}}$ dioperad is the operadic framework.

7.3 Vol I base

Ordered bar complex $B^{\text{ord}}(A) = T^c(s^{-1}\bar{A})$ on $\overline{M}_{g,n}$; averaging map $\text{av}: \mathfrak{g}^{E_1} \rightarrow \mathfrak{g}^{\text{mod}}$ to the symmetric chiral sector.

7.4 Unification via factorisation homology

The three volumes are three *projections* of one underlying factorisation-homology datum:

$$\begin{aligned} \text{Vol I} &= \int_{\overline{M}_{g,n}} \mathcal{F}_C^{E_1} \quad (\text{modular factorisation}), \\ \text{Vol II} &= \int_{C \times \mathbb{R}} \mathcal{F}^{\text{SC}^{\text{ch}, \text{top}}} \quad (\text{HT bulk-boundary}), \\ \text{Vol III} &= \int_{\Sigma_{d-1}} \mathcal{F}_X^{E_d}|_C \quad (\text{CY specialisation}). \end{aligned}$$

All three $\mathcal{F}$'s are instances of one mechanism: a local, holomorphic or holomorphic-topological, factorisation algebra on a complex manifold, integrated against a cycle or modular cycle to produce the projected datum. The CFG E_3 -algebra of 6d hCS observables is the *maximal* instance at $d = 3$, carrying all three projections as specialisations.

7.5 Is the unification load-bearing?

Attack 7.1. The claim “three volumes are three projections of one datum” is aesthetically pleasing but may be a retrofit. What theorem does it enable that is not already stated volume-wise?

Heal 7.2. Two consequences that depend on the unification, not on any one volume:

- (a) The B-row $K^{\kappa_{\text{ch}}} = 8$ identity is cross-volume: Vol I places the ceiling, Vol III supplies the $\mathbf{H}_{\Delta_5}$ -chamber witness, the bridge is the Mukai doubling $24 \rightarrow 8$ (entry $24 \in \{2, 3, 5, 24\}$ maps to entry $8 \in \{0, 8, 13, 250/3, 98/3\}$). This is a genuinely cross-volume theorem; neither volume alone states it.
- (b) The three-sector bar-complex $B_{E_1}/B_{E_2}/B_{E_\infty}$ is a Vol III reading of the Vol II two-coloured $\text{SC}^{\text{ch}, \text{top}}$; the Vol I shadow tower is the B_{E_∞} -symmetric closed sector. The “total shadow obstruction Θ_A ” of Vol I is realised on the $\text{SC}^{\text{ch}, \text{top}}$ -closed-colour component of the Φ_d -image. This is a cross-volume identification that promotes three separate conjectures to one theorem.

The unification is load-bearing at (a) and (b); elsewhere it is clarifying but not theorem-generating.

8 Open questions surfaced

1. (Vol I / Vol III) Curved bar-cobar Quillen equivalence chiral instantiation at compact CY_d with $\kappa_{\text{ch}} \neq 0$: Booth–Lazarev 2304.08409 handles the abstract case; the chiral analogue on $\overline{M}_{g,n}$ awaits.
2. (Vol II) $\text{SC}^{\text{ch}, \text{top}}$ -three-sector stratification $(B_{E_1}, B_{E_2}, B_{E_\infty})$ as a faithful decomposition: state as a chain-level equivalence of filtered bar complexes.
3. (Vol III) Explicit $\text{Sp}_{K3,E}(\Phi_3^{\text{FA}}(K3 \times E)) = U^{\text{ch}}(\mathfrak{heis}_{\text{Muk}}) \otimes U^{\text{ch}}(\mathfrak{g}_{K3}^{\text{BPS}})$ at chain level, closing stratum S_3 of CY- A_3 .
4. (Vol III) Lorgat 2020 Conjecture 1 sharpened: chiral-bialgebra isomorphism $\text{Sp}_{K3,E}(\Phi_3^{\text{FA}}) \cong \mathbf{H}_{\Delta_5}$ with character $= \Delta_5^{-2}$.
5. (Vol III) Fake Monster at $d = 5$: explicit $\text{Sp}_{K3^2,E}(\Phi_5^{\text{FA}}(K3 \times K3 \times E))$ with $\kappa_{\text{BKM}}(\Phi_{\text{FM}}) = 12$.
6. (Vol III) Non-CHL $N = 7$ sibling: order-4 gerbe Cheeger–Simons character realisation; currently obstructed (Nikulin fixed-count 3 forces singular limit).

9 In one line

The trilogy’s content is one mechanism — a local holomorphic (or holomorphic-topological) factorisation algebra $\mathcal{F}$ on a complex manifold, projected by integration against three different cycle data: Vol I integrates against the modular tower $\overline{M}_{g,n}$ to obtain the ordered bar complex with five-theorem A/B/C/D/H content; Vol II integrates against the $C \times \mathbb{R}$ dioperad to obtain $\text{SC}^{\text{ch}, \text{top}}$ and 3D HT QFT bulk-boundary; Vol III integrates against (Σ_{d-1}, C) -cycles on Calabi–Yau d -folds to obtain Φ -image E_1 -chiral shadows specialising to $\mathfrak{g}_{\Delta_5}$, Borchers Monster, Fake Monster, and their Niemeier–CHL cousins — one projection per volume, one functor, the five theorems shared.

A Rectification log

A.1 Sources consulted

- CLAUDE manifestos: Vol I (`~/chiral-bar-cobar/CLAUDE.md`, 463 lines); Vol II (`~/chiral-bar-cobar-vol2/` 366 lines); Vol III (embedded in user instructions, copy of `CLAUDE.md`).
- Vol III `working_notes.tex` (23 000 lines): `grep` audit for Theorem A/B/C/D/H; CY-A/B/C/D; `kappa_ch(K3 x E)`; `c_N(0)`; `two-stage`; `Positselski`; `Hochschild concentration`.
- `notes/platonic_synthesis_waves_11_through_16.tex` (584 lines): ten theorems, one corollary, one conjecture, one retraction table.
- `notes/CoHA_to_W_infty_treatise.tex` (821 lines): three worked examples; CoHA treatise rows.
- `notes/wave12_*.tex`: sixty-seven files (including `wave12_b2_cross_volume_consistency.tex`, `wave12_u1_two_stage_functor.tex`, `wave12_a8_en_hierarchy_kapranov.tex`, `wave12_a11_coha_y_plus_vs_w`, `wave12_a12_six_routes_k3_e.tex`, `wave12_a13_costello_5_routes_6d_hCS.tex`, `wave12_f1_6d_hcs_e3_gener`, `wave12_f4_coha_w_infty_consistency.tex`).

A.2 Attack-heal pass record

Five cycles executed: Stage 1 canonicity under Kontsevich–Tamarkin gap; Stage 2 uniqueness vs genuine moduli on non-product CY_3 ; Dunn–Lurie $\int_{\Sigma_2} E_3 \simeq E_1$ real-vs-complex dimension slip; two-stage extension to $d \neq 3$; falsifiability.

A.3 Out-of-scope (not addressed here)

- Vol II `working_notes.tex` / `working_notes.pdf`: not included in the `grep` audit (outside 800-word scope).
- Vol I `landscape_census.tex` full line-by-line audit: not performed; relied on the Vol II consistency-audit summary at `wave12_b2:45–63`.
- κ_{ch} -value triplet $\{0, 3, 5\}$ at $K3 \times E$: resolution stated in §4 entry 1; physical manuscript rectification (normalising subscripts across sixty-seven `wave12` notes) is deferred to HZ-7 pass at the reader-facing `chapters/` tier.

A.4 Compliance notes

No AI attribution; author Raeez Lorgat. No edits to `main.tex`, `working_notes.tex`, `CLAUDE.md` (any volume), or `chapters/*.tex`. No `make fast`, no commits. Chriss–Ginzburg voice; section titles name mathematical objects; subscript discipline on $\kappa_\bullet$ everywhere; bare κ avoided.